

- 2 A student investigates light from different galaxies.

Fig. 2.1 shows the lines in the absorption spectrum from a distant galaxy.

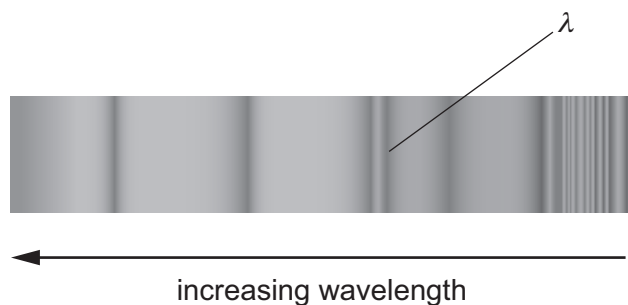


Fig. 2.1

The wavelength of one of the lines in the absorption spectrum is λ . The wavelength of this spectral line in the laboratory is λ_0 .

The observations of the same spectral line are repeated for different galaxies.

The student determines the distance d of each galaxy from the Earth.

It is suggested that λ and d are related by the equation

$$\frac{\lambda - \lambda_0}{\lambda_0} = \frac{Hd}{c}$$

where c is the speed of light in free space and H is the Hubble constant.

- (a) A graph is plotted of λ on the y -axis against $\frac{d}{c}$ on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]

